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Real-Time Facial Emotion Recognition System Using Machine Learning

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INTRODUCTION



◆ Overview of the Project

The project focuses on developing a Real-Time Facial Emotion Recognition System using Deep Learning techniques. It captures live video through a webcam, detects human faces, and classifies emotions such as angry, disgust, fear, happy, neutral, sad, and surprise using a Convolutional Neural Network (CNN). Additionally, the system is designed to continuously capture and store emotional data throughout the day for further analysis.

◆ Motivation / Background

Understanding human emotions plays a crucial role in improving communication, learning environments, and mental well-being. In settings like classrooms or workplaces, it is difficult for teachers or supervisors to manually monitor the emotional state of individuals in real time. This creates a need for an automated, intelligent system that can accurately analyze and track emotions without human intervention.

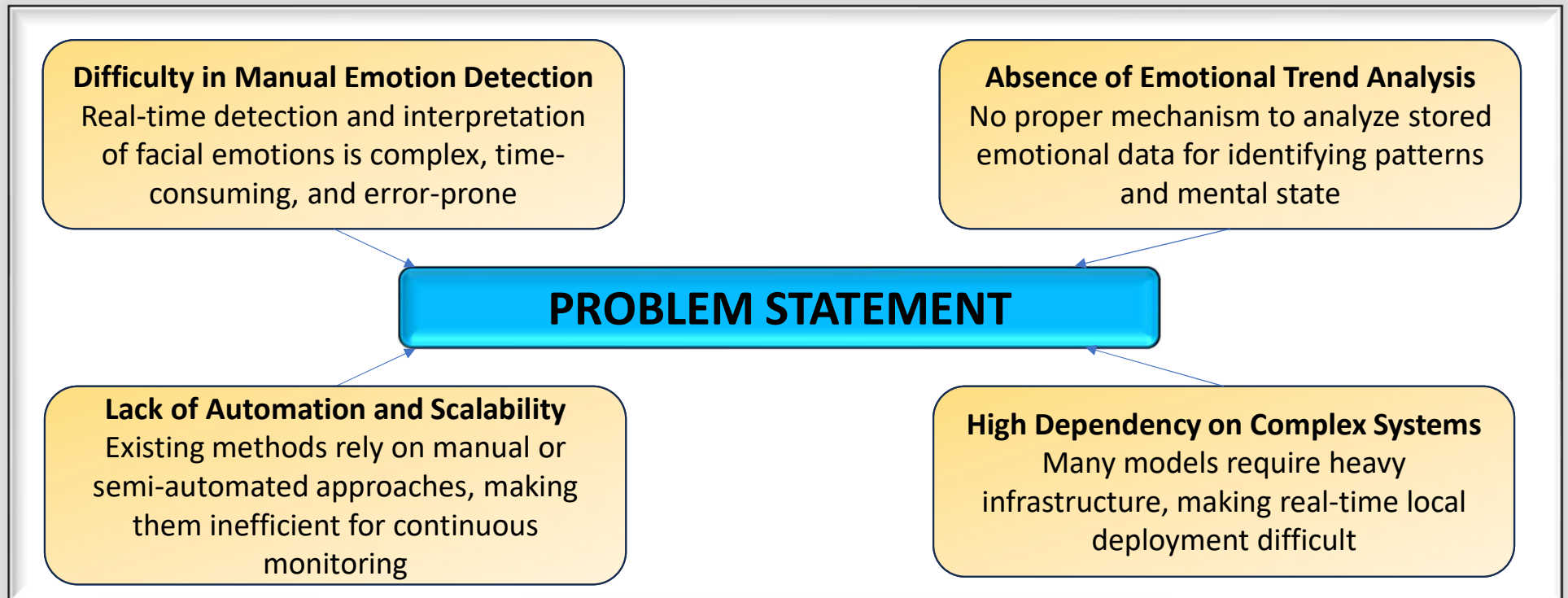
◆ Problem Statement

Manual detection and interpretation of facial emotions in real-time scenarios is challenging, time-consuming, and often inaccurate. There is a need for a system that can not only detect emotions in real time but also continuously monitor, capture, and store emotional patterns throughout the day to help in tracking mental health, engagement levels, and behavioral trends.

PROBLEM IDENTIFICATION



The problem mainly lies in the difficulty of detecting and interpreting human emotions manually in real-time scenarios. Existing methods are time-consuming, less accurate, and lack automation. Moreover, current systems do not continuously capture, monitor, or store emotional data throughout the day, making it difficult to track mental health patterns and emotional trends effectively.



OBJECTIVES



Primary and Secondary Objectives

☐ Primary Objective:

To develop a machine learning-based real-time facial emotion recognition system that can detect human faces and classify emotions such as angry, disgust, fear, happy, neutral, sad, and surprise using a CNN-based model with high accuracy.

☐ Secondary Objectives:

- To implement face detection and preprocessing using OpenCV techniques
- To utilize a multi-layer Convolutional Neural Network (CNN) for efficient feature extraction and classification
- To enable real-time emotion detection using webcam input
- To continuously capture, monitor, and store emotional data throughout the day
- To analyze emotional patterns over time for tracking engagement and mental state
- To improve system performance using data augmentation and optimization techniques

Scope of the Project

The system is designed to detect and classify facial emotions and also maintain a record of emotions throughout the day for trend analysis. However, the scope is limited to facial expressions only and does not include other behavioral factors such as voice, body language, or eye tracking. Large-scale deployment and full classroom integration are beyond the current scope.

LITERATURE REVIEW / EXISTING SYSTEM



Summary of Existing Solutions or Related Work

Traditional facial emotion recognition methods such as LBP, HOG, and SVM provided limited accuracy due to manual feature extraction. With advancements in deep learning, CNN-based models like VGG, ResNet, and MobileNet significantly improved performance by automatically learning features from images. Furthermore, the use of transfer learning techniques, especially MobileNetV2, has enabled more efficient and faster real-time emotion detection systems.

Gaps Identified

While the reviewed research showed promising results, several important gaps still remain:

- Limited real-time implementation in practical environments like classrooms.
- Lack of true real-time monitoring capability
- Identification limited to only one factor (e.g., only eye contact or only lip movement)
- Use of high-complexity deep learning architectures that are not suitable for normal desktop system
- Most systems do not store or track emotions over time for mental health analysis

These limitations highlight the need for an automated, real-time system capable of accurately detecting, monitoring, and tracking emotions over time.

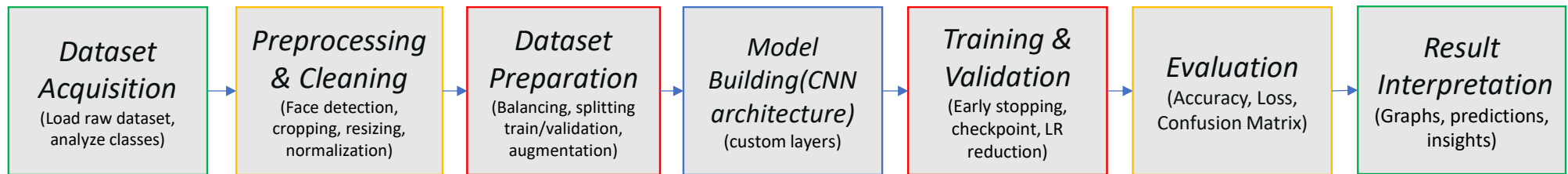
PROPOSED METHODOLOGY



Research Design

The project follows a Supervised machine learning approach to detect and classify facial emotions into seven categories using a CNN model, with real-time monitoring and emotion tracking.

The design includes:



STEP 1 : Data Collection Method

The data was collected from a publicly available dataset:

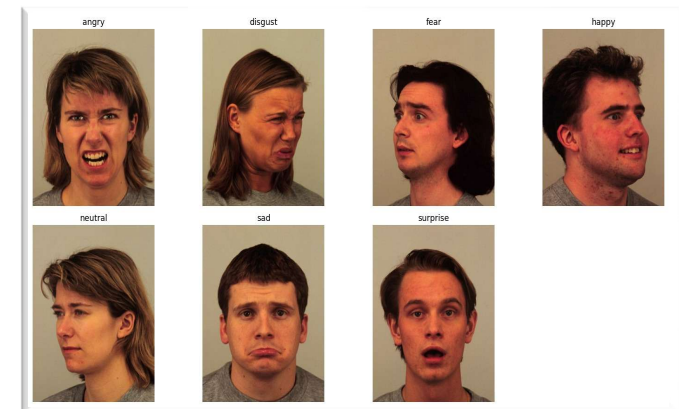
Dataset: Face Emotion Detection Dataset

Classes: angry, disgust, fear, happy, neutral, sad, surprise

(Each class contains 400+ images)

Source: Kaggle

URL: <https://www.kaggle.com/datasets/biswas2022/face-emotion-detection/settings>



STEP 2 : Data Preprocessing and Augmentation



Preprocessing

- **Image Resizing** : All images are resized to a fixed dimension (96×96) to ensure uniform input for the model.
- **Normalization** : Pixel values are scaled from 0–255 to 0–1 to improve model performance and faster convergence.
- **Dataset Splitting** : The dataset is divided into training and validation sets to evaluate model accuracy.

Augmentation

- **Rotation & Zoom** : Images are rotated and zoomed to handle different angles and scales.
- **Horizontal Flip** : Flipping increases data variation and improves generalization.
- **Reduce Overfitting** : Helps the model learn better and perform well on unseen data.

STEP 3 : Model Building (CNN) and Training

Building

- **CNN Architecture** : Custom CNN with 3 Conv + MaxPooling layers (32, 64, 128 filters) .
- **Feature & Classification Layers** : Flatten layer with Dense (128, ReLU) and Dropout (0.4).
- **Output & Optimization** : Softmax layer (7 classes) with Adam optimizer and EarlyStopping.

Training

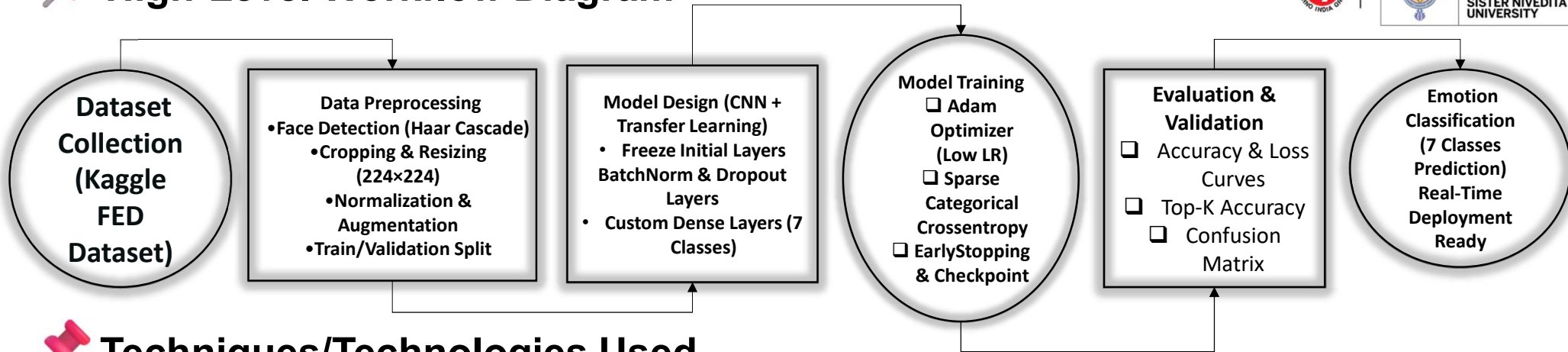
- **Loss Function** : Categorical Crossentropy for multi-class classification.
- **Optimizer & Metrics** : Adam optimizer with accuracy as evaluation metric.
- **Training Process** : Model trained for approximately 100–150 epochs.

STEP 4 : Model Evaluation & Saving

- **Performance Evaluation** : Generated **accuracy & loss graphs** and used **confusion matrix** to analyze model performance .
- **Model Saving** : Saved the best-performing model (*best_model.h5*) for real-time prediction.



High-Level Workflow Diagram



Techniques/Technologies Used

- **Programming & Model** : Python, CNN (Convolutional Neural Network), Transfer Learning
- **Libraries** : TensorFlow/Keras, OpenCV, NumPy, Matplotlib
- **Platform** : Google Colab (training), VS Code (deployment)

CHALLENGES FACED

- **Overfitting** : High training accuracy but lower validation accuracy.
- **Misclassification** : Confusion between emotions like Neutral, Fear, and Surprise .
- **Real-Time Prediction Issue** : Model performed well on images but less accurate on webcam .
- **Lighting & Detection Issues** : Poor lighting and face detection affected prediction quality.

SOLUTIONS IMPLEMENTED

- ✓ **Overfitting Solution** : Used Dropout layers, Data Augmentation, and optimized learning rate .
- ✓ **Misclassification Fix** : Analyzed confusion matrix, balanced dataset, and improved preprocessing .
- ✓ **Real-Time Issue Fix** : Ensured same preprocessing as training (resize 96x96, normalization) for accurate webcam prediction.
- ✓ **Lighting Problem Fix** : Improved lighting conditions and accurate face cropping.

IMPLEMENTATION

● Model Development, Training & Evaluation

CNN Model Development

- Built a CNN using Conv2D and MaxPooling layers for feature extraction
- Used Dense layers for classification of 7 emotion classes,
- Applied Dropout and regularization to reduce overfitting.

Model Training

- Trained the model on training and validation datasets
- Epochs: 120, Batch size: 32
- Used Adam optimizer and monitored training & validation accuracy

Model Evaluation

- Evaluated performance using accuracy and loss curves
- Used Top-K accuracy to check if correct label is within top predictions
- Analyzed model performance for reliability and generalization



Custom CNN Architecture

- 3 Conv layers (32, 64, 128) for feature extraction
- MaxPooling for dimensionality reduction
- Dense + Dropout (0.4) to reduce overfitting
- Softmax layer for 7-class emotion classification

```
model = models.Sequential([
    layers.Input(shape=(IMG_SIZE, IMG_SIZE, 3)),

    layers.Conv2D(32, 3, padding='same', activation='relu'),
    layers.MaxPooling2D(),

    layers.Conv2D(64, 3, padding='same', activation='relu'),
    layers.MaxPooling2D(),

    layers.Conv2D(128, 3, padding='same', activation='relu'),
    layers.MaxPooling2D(),

    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dropout(0.4),

    layers.Dense(NUM_CLASSES, activation='softmax')
])
```

OUTCOME

 **Expected Results:** Based on the experiments conducted:

Final Training Accuracy : 0.9123

Final Validation Accuracy : 0.840

 **TRAIN ACCURACY**

Top-2 Accuracy (Train): 99.32%

Top-3 Accuracy (Train): 99.91%

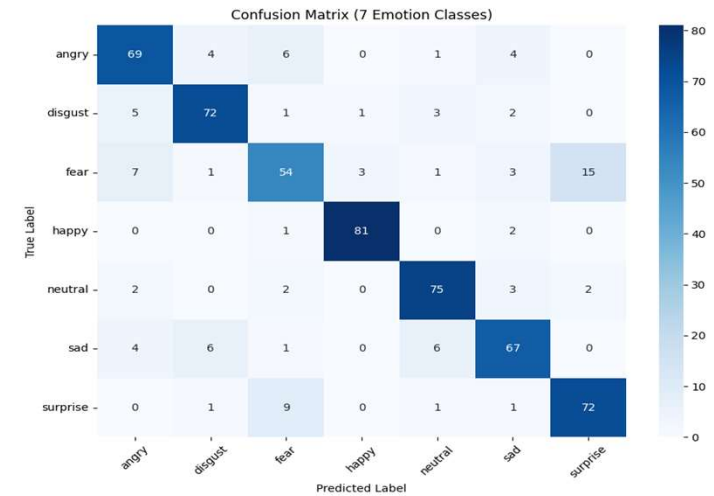
Expected trends:

- *Data augmentation and preprocessing improve model generalization and reduce overfitting.*
- *The model shows stable learning with consistent accuracy after sufficient training epochs.*

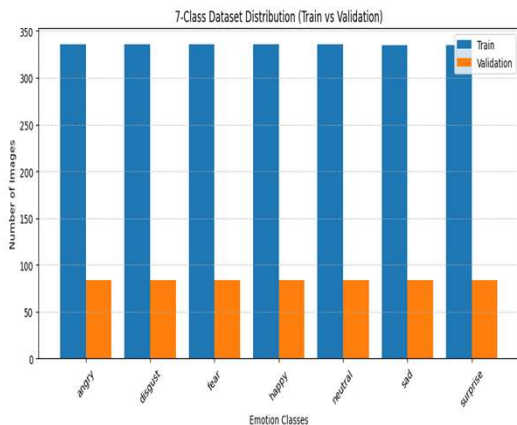
 **VALIDATION ACCURACY**

Top-2 Accuracy (Val): 95.41%

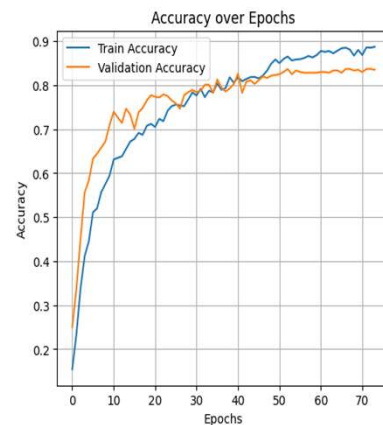
Top-3 Accuracy (Val): 98.47%



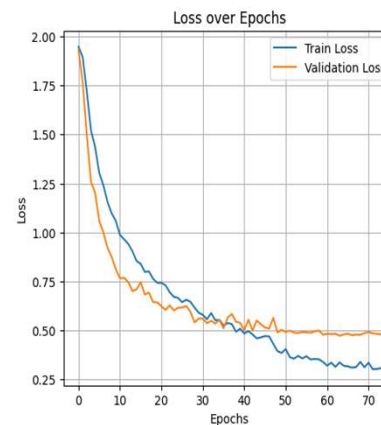
Confusion Matrix



Train Validation Distribution



Accuracy Loss Graph

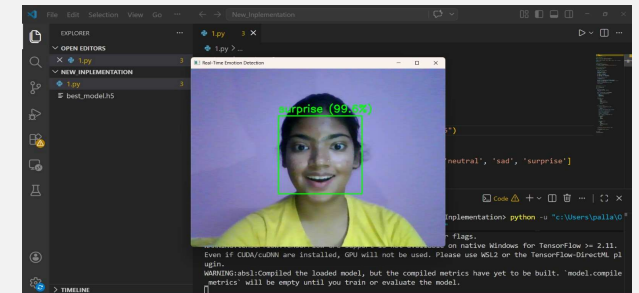
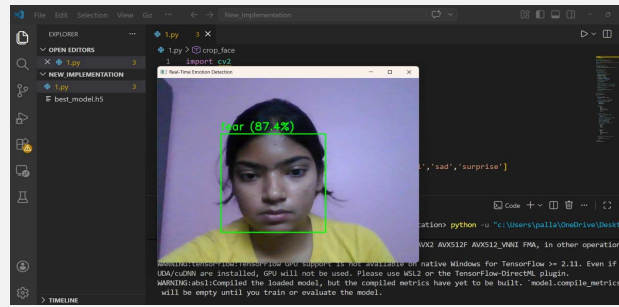
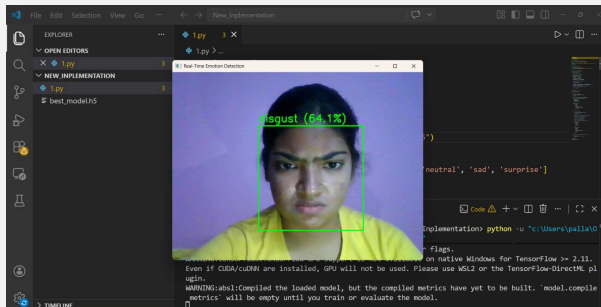
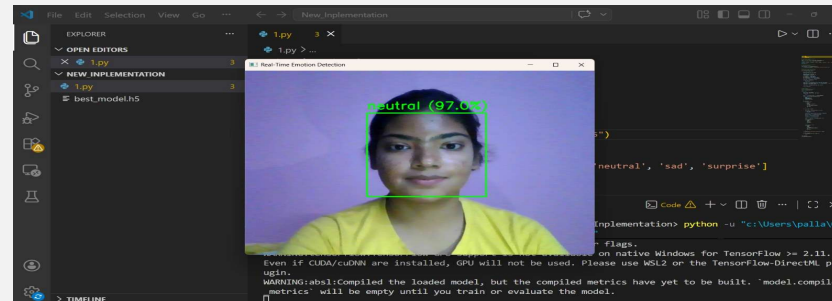
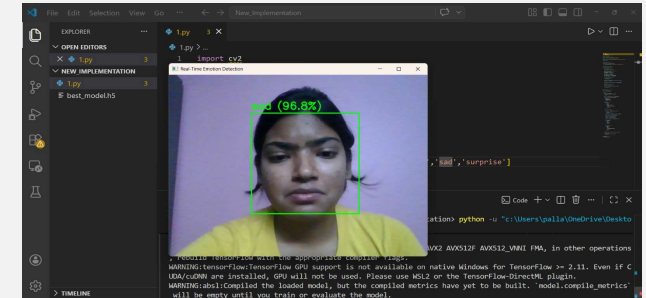
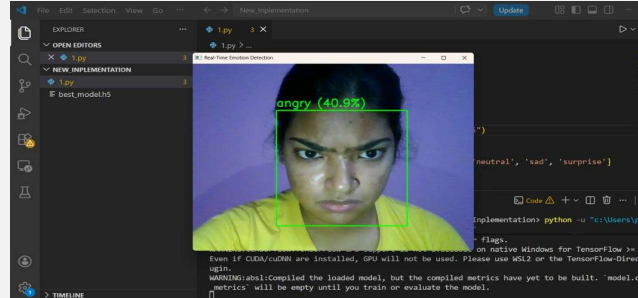
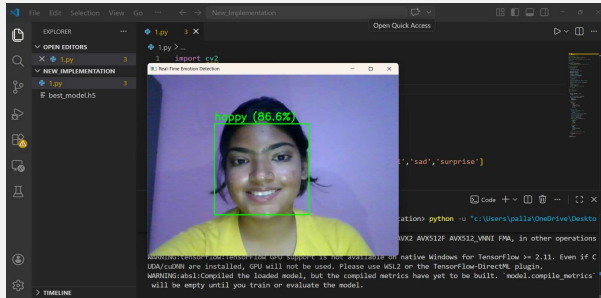


*** Classification Report:

	precision	recall	f1-score	support
angry	0.8022	0.8690	0.8343	84
disgust	0.8506	0.8810	0.8655	84
fear	0.7403	0.6786	0.7081	84
happy	0.9639	0.9524	0.9581	84
neutral	0.8902	0.8690	0.8795	84
sad	0.8025	0.7738	0.7879	84
surprise	0.8046	0.8333	0.8187	84
accuracy			0.8367	588
macro avg	0.8363	0.8367	0.8360	588
weighted avg	0.8363	0.8367	0.8360	588

Classification Report

Real-Time Emotion Detection: Live Webcam Output



CONCLUSION



SUMMARY OF KEY POINTS

Successfully developed a real-time facial emotion recognition system using CNN architecture

The model effectively detects and classifies 7 different emotions from facial expressions

Applied preprocessing, data augmentation, and transfer learning to improve accuracy and performance

Implemented evaluation techniques such as accuracy, loss graphs, and confusion matrix for validation

Enabled continuous monitoring and storage of emotional data to analyze patterns over time

Overall, the system provides an automated, efficient, and scalable solution for emotion analysis



LIMITATIONS



Despite achieving satisfactory performance, the proposed system has certain limitations :-

- Model performance may be affected by variations in lighting conditions, facial orientation, and occlusion
- Dataset may not fully represent real-world diversity, affecting generalization
- System currently supports single-face detection and is limited in multi-person scenarios
- Real-time performance depends on system hardware and processing capability
- Similar emotions (e.g., fear and surprise) may be misclassified due to overlapping features
- Continuous emotion tracking is basic and lacks advanced analytical insights



FUTURE WORKS

- **Integration with Video Conferencing**
Can be integrated with platforms like Google Meet or Zoom for real-time emotion analysis
- **Participant Engagement Monitoring**
Helps track attention levels and provide feedback during meetings
- **Background Emotion Detection System**
Develop a desktop application for Windows/Linux that runs automatically
- **Real-Time Continuous Monitoring**
System runs in background, accesses webcam (with permission), and detects emotions in real time

This work highlights the potential of intelligent systems while opening paths for further advancements in accuracy, scalability, and real-time applications

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